Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 03
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A voice dialog that fails on purpose

What is this sheet about?

This sheet turns the dialog you dissected on paper into a running voice prototype. You build it on one platform of your choice and then provoke the failures on purpose: lax confirmation, ignored barge-in, no escalation to a human. What you document is not the happy path but the moments where the dialog loses its user.

Docking point: This exercise cashes in session T07 (*Dialogs nobody wrote*): the pharmacy dialog you dissected on paper — confirmation logic too lax, barge-in ignored, no escalation to a human — gets rebuilt as a running prototype, and this time you provoke the failures for real.

Organisation: Please work in your project team of 3–4 students. Pick **one** platform per team. *ProtoPie* with its built-in voice prototyping (speech-to-text and text-to-speech triggers) is the right choice if your dialog lives inside a screen prototype. *Voiceflow*, once an Alexa-skill designer and now an LLM-agent platform, is the right choice for a standalone conversational agent. The platform's own transformation is course material: the dialog tree you draw and the model behavior you configure are two different design objects, and you will feel the difference today. Headphones are recommended: six teams talking to six prototypes in one room is itself a lesson in modality conflict.

AI role — tool and object of examination (both declared): Tool, because you build with platforms that generate dialog behavior; object of examination, because that generated behavior is exactly what you attack in the third assignment. Your upload must disclose what was made with which tool.

Bonus points: In order for the bonus rule to apply (participation in all six blocks with an upload within 48 hours each earns 5 % compensation on the exam), your team must hand in one complete submission package (see the end of this sheet) within 48 hours after this exercise. This is block three of six, the halfway point.

First assignment (15 minutes): Scope the dialog (team). One task from your project idea (or the T07 pharmacy refill, if your project has no natural dialog task), three to six turns, written as the ideal transcript (the *happy path*), **before you touch a tool**. If you can, spend five of these minutes on a human rehearsal: one team member wants the task done, another plays the system with the real data in front of them. Voice only, backs turned, audio recorded with consent. Write the transcript from what was actually said. Mark the one step where a recognition error would hurt most; for the pharmacy case that is the medication confirmation, for your case, find its equivalent.

Second assignment (45 minutes): Build the happy path (team, split roles). Get the ideal transcript running end to end. Builders build; **one team member stays off the tool** and writes the state model of what is being built. You will need it in the fourth assignment, and the divergence between the two is usually where the failure hides.

Third assignment (35 minutes): Provoke the failure paths (pairs, roles swap halfway). One person operates, one sabotages from a written attack list. At minimum:

- interrupt mid-prompt (barge-in)
- mumble

- stay silent
- give an ambiguous answer at the critical step
- ask something off-task
- try to reach a human
- take a shortcut: say everything the dialog needs in one sentence, as an expert would

Log every attack: what you did, what the system did, expected vs. actual. **Do not fix anything yet: the one repair of this block comes in the fifth assignment.**

Fourth assignment (25 minutes): Document one failure completely (team). Choose **the failure you learn the most from** and write it up: transcript, position in the state model, cause (recognition, dialog logic, or generated-behavior drift), and the T07 principle it violates. For a recognition-side cause, name the layer: *no speech detected*, *speech not recognized*, or *wrong recognition*; words that arrived correctly but found no path in the dialog (*recognized, not handled*) are a dialog-logic cause. For the violated principle, use the eight voice rules as a checklist: quality, quantity, relevance, manner, back-and-forth exchange, no series of on-offs, knowledge of previous interactions, independent completion with an exit to a human. This document is the milestone artifact, and it counts only with the cause explained.

Fifth assignment (20 minutes): Build one guardrail and retest (team). Close exactly one hole: an explicit confirmation, a re-prompt strategy, an escalation to a human. Rerun the attack that found it, and record whether the guardrail holds and what it costs (usually: added turns, added friction; T02's price tag applies to repairs too).

Sixth assignment (10 minutes): Upload preparation. Package while the transcripts are fresh.

Buffer and critique (30 minutes): Failure gallery: each team plays or reads its documented failure in two minutes; the room answers one question: *would a first-time user have recovered?* The buffer also absorbs platform friction, which with voice tooling is not hypothetical.

The submission (within 48 hours, one upload per team, via Moodle) consists of:

- `dialog-prototype` — ProtoPie file or Voiceflow share link, running happy path plus guardrail.
- `failure-case.md` — the fully documented failure: transcript, state-model position, cause, violated principle, guardrail and its price (max 3 pages).
- `attack-log.md` — the full sabotage list with outcomes, one line each (max 1 page).
- A recording of the provoked failure and one of the guardrail holding (max 2 × 90 seconds, .mp4 or .m4a, max 100 MB total).
- **AI disclosure line** (in `failure-case.md`, mandatory): *"Made with AI: [tool] was used for [part]; checked and understood by [name]."*

Put everything into one upload per team and hand it in via Moodle within 48 hours after this exercise.